

Part A – Conceptual Remote Sensing

(27 points total – each question is one point unless otherwise noted)

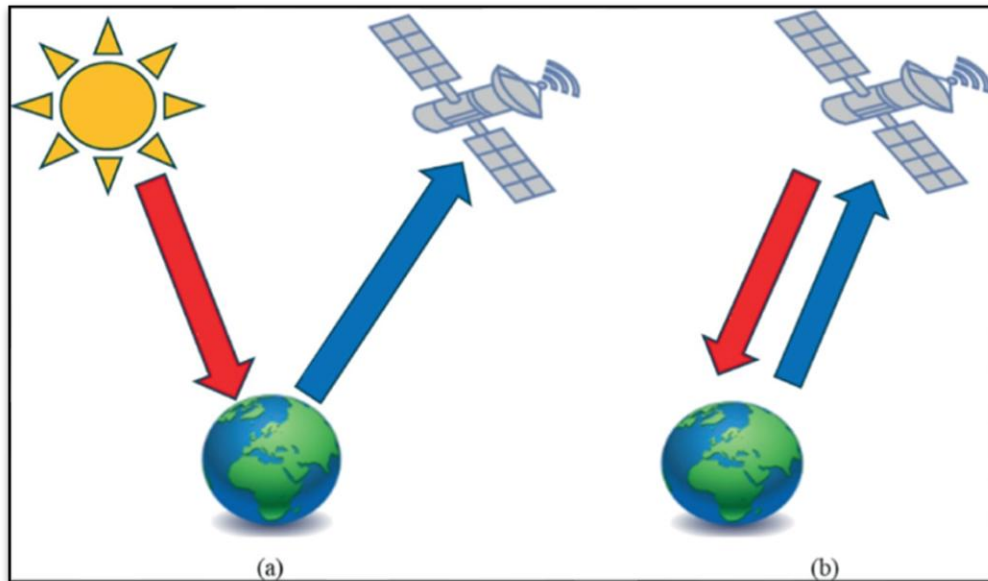
1. As compared to blue light, red light has... [Select all that apply]
 - a. Higher frequency
 - b. Longer periods
 - c. High energy
 - d. Longer waves
 - e. None of the above

2. Unlike radio waves, infrared waves... [Select all that apply]
 - a. Have shorter wavelengths
 - b. Travel faster
 - c. Cannot be reflected – only absorbed
 - d. Are not useful for Earth observation
 - e. None of the above

3. The windows of a building are coated with a substance that reflects *red* light, absorbs *green* light, and transmits *blue* light. When the building's lights are off during the day, what color do the windows appear?
 - a. Green from outside; Red from the inside
 - b. Red from the outside; Yellow from the inside
 - c. Green from the outside; Yellow from the inside
 - d. Red from the outside; Blue from the inside

4. The sky appears mainly blue on a clear, sunny day because
 - a. Red wavelengths of sunlight are absorbed by gasses in the air
 - b. Blue wavelengths of sunlight are absorbed by large particles in the air
 - c. Blue wavelengths of sunlight are scattered by gasses in the air
 - d. Red wavelengths of sunlight are scattered by large particles in the air

5. Solar irradiance is the amount of sunlight energy _____ by Earth per unit of time.
 - a. Emitted
 - b. Received
 - c. Reflected
 - d. Absorbed
 - e. Transmitted

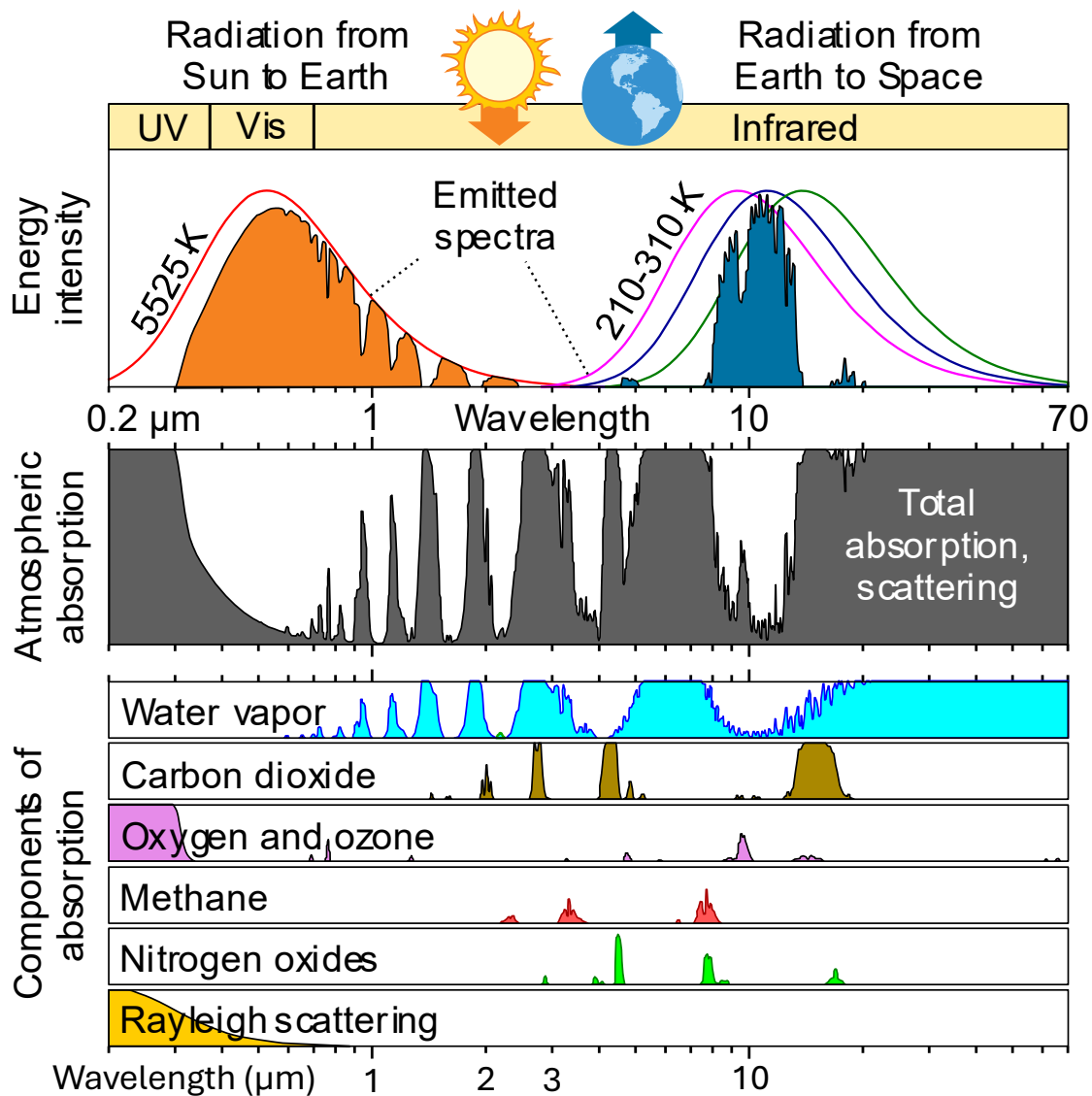


6. Which diagram in the picture above shows *passive* remote sensing?
 - a. **A**
 - b. B
 - c. Both
 - d. Neither

7. Which diagram in the picture above shows a remote sensing system that could collect altimetry data?
 - a. A
 - b. **B**
 - c. Both
 - d. Neither

8. A university research group is developing a remote sensing mission that can detect forest fires in otherwise inaccessible locations. What wavelength band should the sensor be sensitive to in order to detect the heat signature of a fire?
 - a. **Infrared**
 - b. Radio
 - c. Ultraviolet
 - d. Microwave
 - e. Visible

9. Which type of sensor would the sensor from Question 8 be?
 - a. Active
 - b. **Passive**



10. Earth primarily emits radiation in the _____ portion of the spectrum.
infrared

11. Portions of the spectrum such as near wavelengths such as 10 μm (micrometers) where absorption and scattering of wavelengths is low are known as _____.

Atmospheric windows (just “windows” o.k.)

12. Gases such as water vapor, carbon dioxide, ozone, and methane are known as _____ due to their ability to absorb and re-radiate energy emitted by the Earth. Greenhouse gases

13. (3 pts) Would a sensor that is sensitive to wavelengths between 2.5 and 3.0 μm be useful on an Earth-observing satellite? Why or why not?

It would not (+1) because those wavelengths are mostly absorbed and scattered in the atmosphere (aren't transmitted) (+1.5), especially by water vapor and carbon dioxide (+0.5).

14. A weather-observation remote sensing mission is being designed and will need to provide updated imagery for a location at least every 10 minutes. Which type of orbit would be ideal for the satellite carrying the sensors?
- a. Low-Earth orbit
 - b. Medium-Earth orbit
 - c. High-Earth orbit
 - d. **Geostationary orbit**
 - e. All of the above
15. Which type of resolution is the satellite from the previous question attempting to maximize?
- a. Spectral resolution
 - b. Spatial resolution
 - c. **Temporal resolution**
 - d. Radiometric resolution
 - e. All of the above
16. A satellite company is designing a remote sensing mission that can provide data about land surfaces. They hope to produce imagery with the finest detail possible in order to be able to detect small objects within the images, ideally objects less than 2 meters long. Which type of orbit would be ideal for the satellite carrying the sensors?
- a. **Low-Earth orbit**
 - b. Medium-Earth orbit
 - c. High-Earth orbit
 - d. Geostationary orbit
 - e. All of the above
17. Which type of resolution is the satellite from the previous question attempting to maximize?
- a. Spectral resolution
 - b. **Spatial resolution**
 - c. Temporal resolution
 - d. Radiometric resolution
 - e. All of the above
18. Many Earth-observation satellites use a type of low-Earth orbit called a polar or near-polar orbit because they...
- a. Are easier to reach orbit than equatorial low-Earth orbits
 - b. **Can provide nearly global coverage**
 - c. Minimize the amount of geometric distortions in imagery
 - d. Enable higher radiometric resolution

19. Key differences between a geosynchronous and a geostationary orbit are... [Select all that apply]

- a. The orbital period of the satellites
- b. The orbital inclination (angle relative to the equator)
- c. Geostationary orbits must be circular
- d. The distance of a geosynchronous orbit is half the distance of a geostationary orbit
- e. None of the above

20. Hyperspectral sensors typically have higher _____ than multispectral sensors.

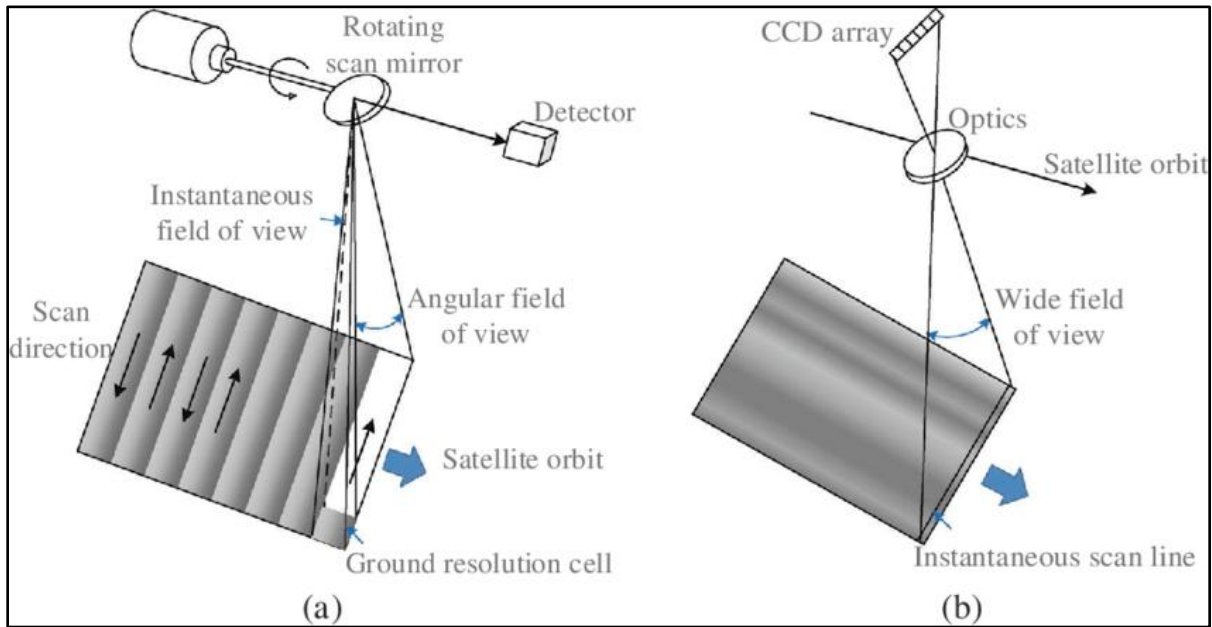
- a. Spectral resolution
- b. Spatial resolution
- c. Temporal resolution
- d. Radiometric resolution
- e. All of the above

21. An important advantage of using hyperspectral sensors rather than multispectral sensors is that they...

- a. Collect less data, reducing technical requirements for data transmission
- b. Can operate at night as well as during the day
- c. Can better tell different ground surfaces or materials apart
- d. Can produce R-G-B false-color composite imagery
- e. All of the above

22. The Operational Land Imager (OLI) sensors aboard Landsat 8 and 9 have 12-bit radiometric resolution, meaning that they can detect...

- a. 12 wavelength bands
- b. 12^2 wavelength bands
- c. 2^{12} wavelength bands
- d. 12 brightness levels
- e. 12^2 brightness levels
- f. 2^{12} brightness levels



23. What is the name for the type of imaging scanner shown in Diagram (a) on the left?

“across-track” OR “whiskbroom” scanner

24. What is the name for the type of imaging scanner shown in Diagram (b) on the right?

“along-track” OR “pushbroom” scanner

25. Which Diagram shows an imaging scanner that would generally have improved spatial and radiometric resolution as compared to the other?

- a. (a)
- b. (b)
- c. Both would allow for roughly equal resolution

Part B – Applied Remote Sensing

(58 Points total – point values provided with question)



Natural-color



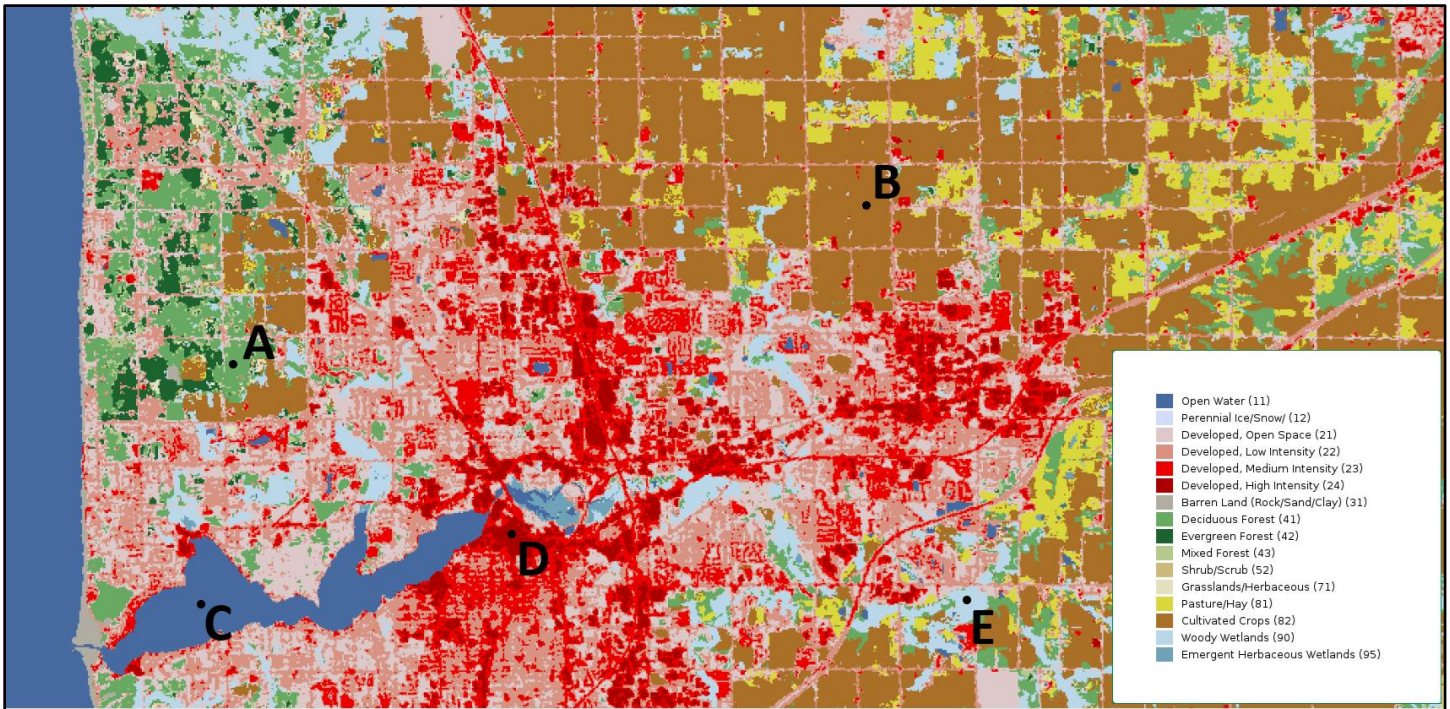
False-color

26. (10 pts) Shown above are two images from data collected by Landsat 8's Operational Land Imager (OLI). The left image is a natural-color image, meaning that the light detected by its red, green, and blue bands are shown as red, green, and blue, respectively.

The right image is a false-color image, created using the same three bands, but re-ordered. Fill in the table below with which band is shown as which color (2.5 pts each). On the blank lines below the table, explain how you arrived at your answers (2.5 pts).

Natural-color image		False-color image	
<i>OLI Band</i>	<i>...is displayed as the color...</i>	<i>OLI Band</i>	<i>...is displayed as the color...</i>
Band 4 (Red)	Red	Band 2 (Blue)	Red
Band 3 (Green)	Green	Band 3 (Green)	Green
Band 2 (Blue)	Blue	Band 4 (Red)	Blue

any response that thoroughly compares the colors between the two images will suffice

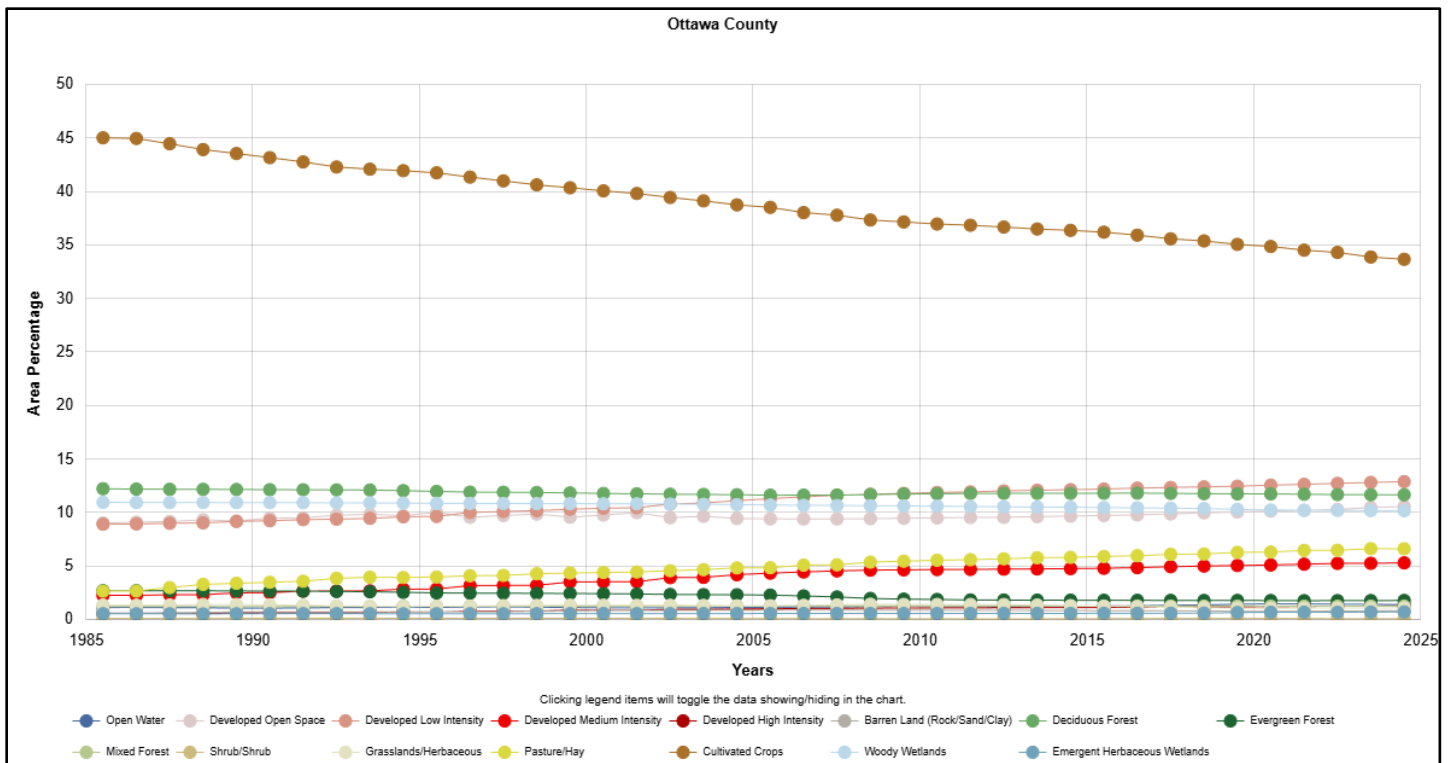


27. (10 pts; 2 pts each) Using the NLCD Land Cover map of the Holland-Zeeland area above, identify which type of land cover is present at each of the five point locations.

- A. _____ deciduous forest
- B. _____ cultivated crops
- C. _____ open water
- D. _____ developed, high-intensity
- E. _____ woody wetlands

28. (2 pts) Which of the sixteen classifications listed in the map's legend would the West Ottawa High School Athletic Fields most likely be identified as?

Developed, open-space



The chart above shows the percentage of Ottawa County land that is classified into each category from the years 1985 through 2024. Use it to answer the following questions.

29. (1 pt) Did the percentage of land classified as “Cultivated Crops” increase or decrease from 1985 to 2024?

decrease

- a. (2 pts) By how much? Show your work.

$34\% (2024) - 45\% (1985) = -11\%$ (of Ottawa Co. land) (negative not needed for points)

- b. (2 pts) What percentage change is this, as a fraction of the original 1985 Cultivated Crop percentage? Show your work.

$(34\% - 45\%) / 45\% = -24\%$ change or 24% decrease (answers 23-26% OK)

30. (1 pt) Did the percentage of land classified as “Developed, Low-Intensity” increase or decrease from 1985 to 2024?

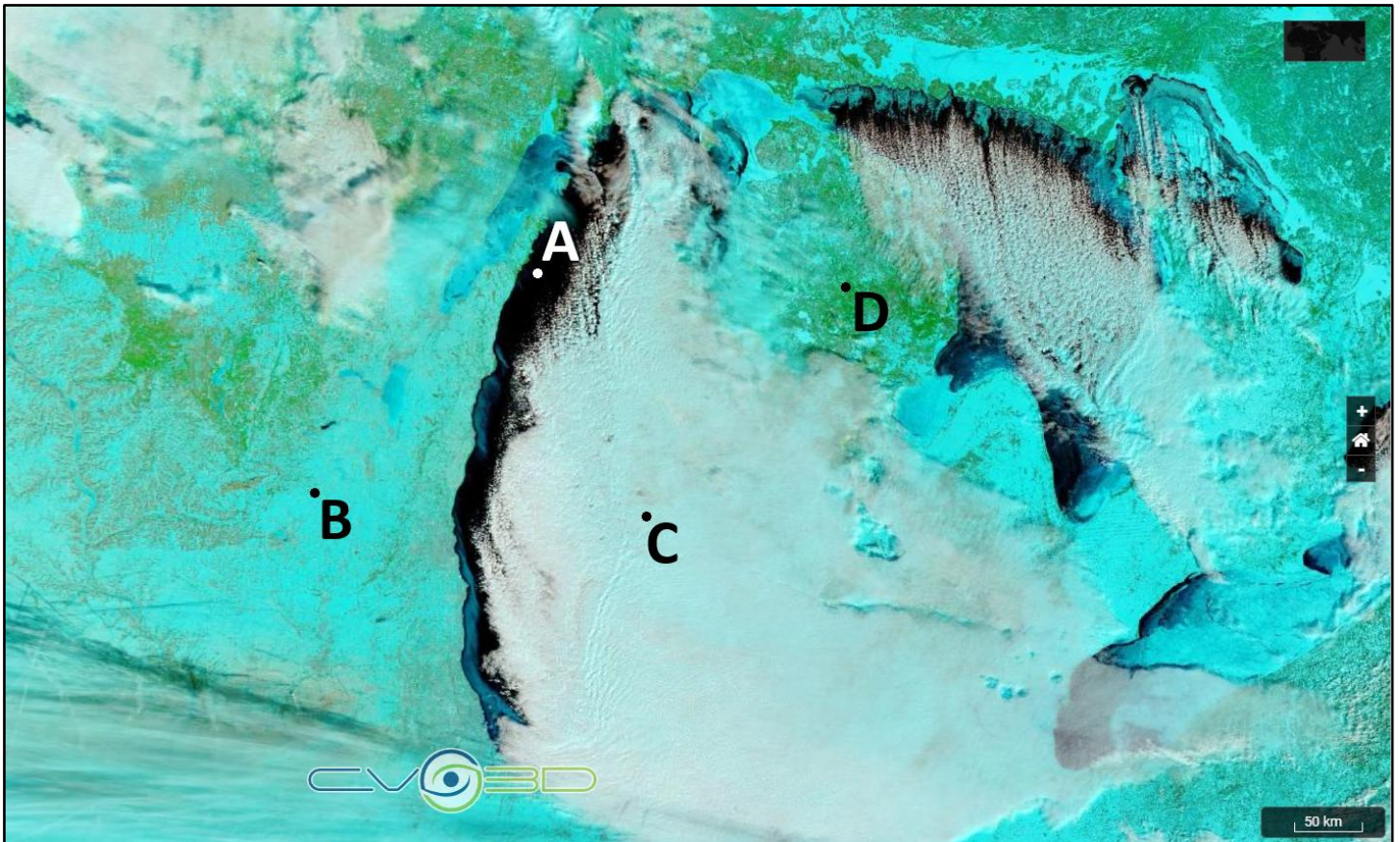
increase

- a. (2 pts) By how much? Show your work.

$13\% (2024) - 9\% (1985) = 4\%$ (of Ottawa county land)

- b. (2 pts) What percentage change is this, as a fraction of the original 1985 Developed, Low-Intensity percentage? Show your work

$(13\% - 9\%) / 9\% = 44\%$ increase (answers 40-50% OK)



The image above shows Michigan's Lower Peninsula and the surrounding area as captured by the MODIS instrument aboard the satellite Terra. It is a false-color image that shows shortwave-IR as red, near-IR as green, and visible red as blue. Use it to answer the following questions.

31. (8 pts; 2 pts each) Identify what is present in the image at the four points causing the respective colors:

A. (black color)

water

B. (light-blue color)

snow/ice

C. (white-ish color)

cloud

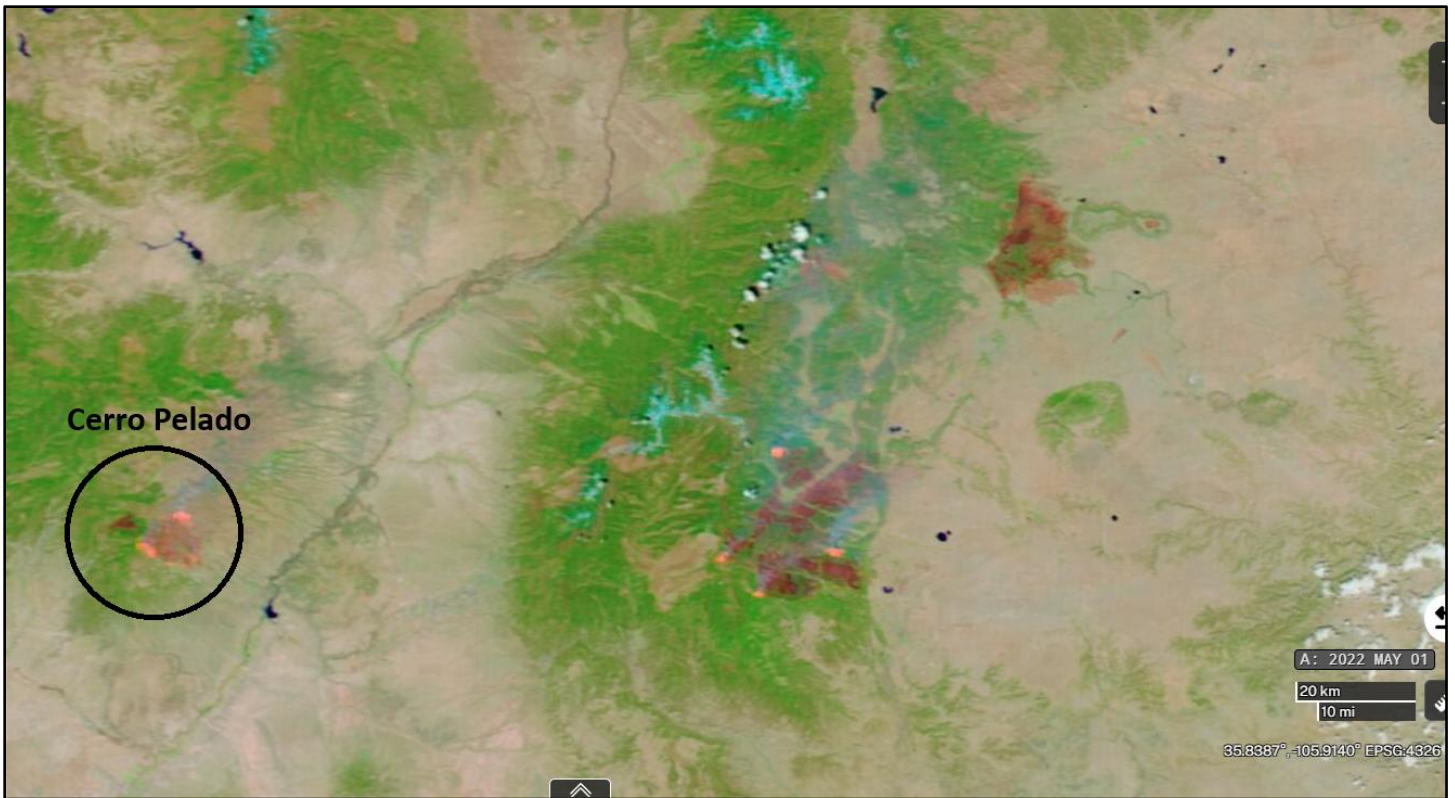
D. (bright green color)

vegetation

32. (1 pt) Is the MODIS sensor active or passive?

passive

Team #: _____



The image above shows forest fires burning in northern New Mexico on May 1st, 2022. It was captured by Terra's MODIS instrument and uses shortwave-IR as red, near-IR as green, and visible red as blue. Use it to answer the following questions.

33. (4 pts) Why is this band combination an effective choice for showing forest fires?

there is a strong contrast between living (green) and burned (reddish-brown) vegetation (+3)
can identify location of active fires (+1)

34. (5pts) What is the area, in square kilometers, of the roughly triangular area burned by the Cerro Pelado fire as of May 1st? Show your work (Area of a triangle = $\frac{1}{2} \times \text{base} \times \text{height}$)

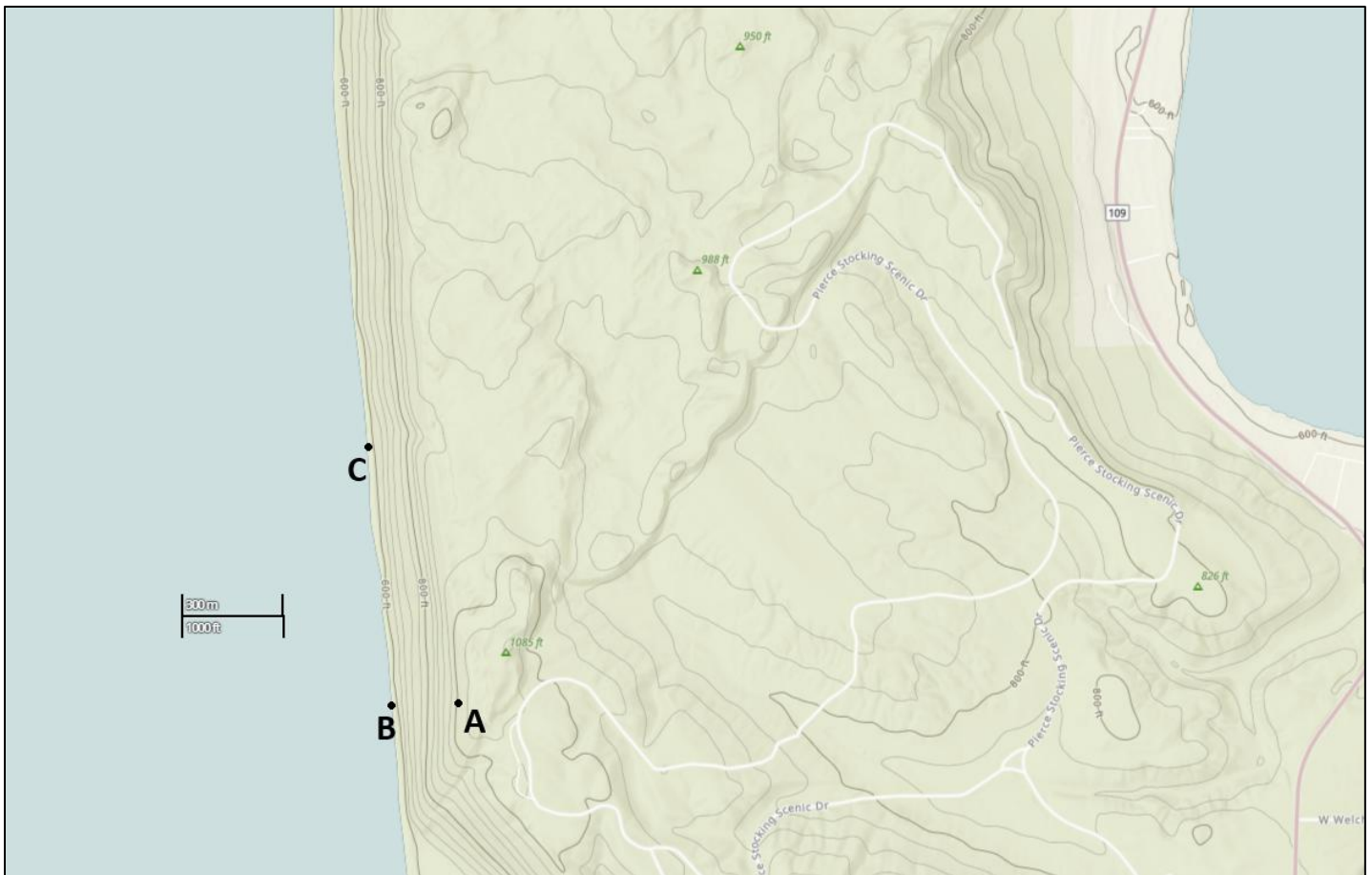
$$A = \frac{1}{2} \times b \times h$$

$$b = 1.1 \text{ cm} \times (20 \text{ km} / 1.6 \text{ cm}) = 13.75 \text{ km}$$

$$h = 0.8 \text{ cm} \times (20 \text{ km} / 1.6 \text{ cm}) = 10 \text{ km}$$

$$A = \frac{1}{2} \times (13.75 \text{ km}) \times (10 \text{ km}) = \text{about } 68.7 \text{ km}^2 \text{ (64-73 km}^2 \text{ OK)}$$

(+1 for showing each dimension scale conversion, +2 for numerical answer 64 to 73, +1 correct unit)



Use the topographic map of Sleeping Bear Dunes above to answer the following questions.

35. (1pt) What is the interval for the thick contour lines?

200 ft

36. (1 pt) What is the interval for the thin contour lines?

40 ft

37. (1 pt) A scenic overlook is located near the highest point on the dunes. What is its elevation?

1085ft

38. (2 pts) Point C is located at an azimuth of 342° relative to Point A. What is this direction expressed as a bearing?

N18° W

39. (3 pts) Which line, AB or AC, has a steeper profile along its length? How can you tell?

AB is steeper (+1)

Both lines experience the same elevation change, but with AB it occurs over a shorter horizontal distance (+2) (only award +1 for saying “the contours are closer together”)